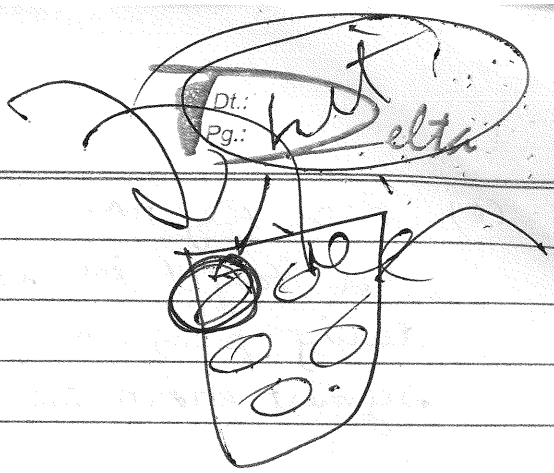


Instruction fetch (IF)

Instruction decode (ID)

operand fetch (OF)

execute (EX)

write back (store)

registers (flip flops)

cache (Static RAM)

main memory (dynamic RAM)

magnetic disk

magnetic tapes

Cache works on LOR (locality of reference)

Cache → is divided in equal parts called block / cache lines

Cache ⟷ data is transferred to cache in form of blocks
thus, main memory & cache is divided into equal size blocks

Cache lines size = main memory block size.

No. of cache lines = $\frac{\text{Cache size}}{\text{block size}}$

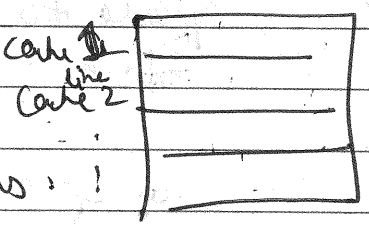
Mapping technique

memory → K mod N = i → cache memory line no, where it should go.
no. of cache line

(i) direct mapping → func^n

← physical address →

tag	Block	word.



Replacement is done by "conflict miss" occurs!!
Hit ratio is less

The delay of tag comparator is called hit latency or hit delay. So, reference is forwarded to main memory.

By using mapping func^n, main memory block is transferred to corresponding cache line along with complete tag. Later CPU accesses ~~to~~ data from cache.

(ii) Associative mapping.

tag	word.

← physical address →

Any main memory block can be mapped to any location in cache. Thus, cache address is not used.

replacement is done only when cache is full.

max no. of tag comparisions = N

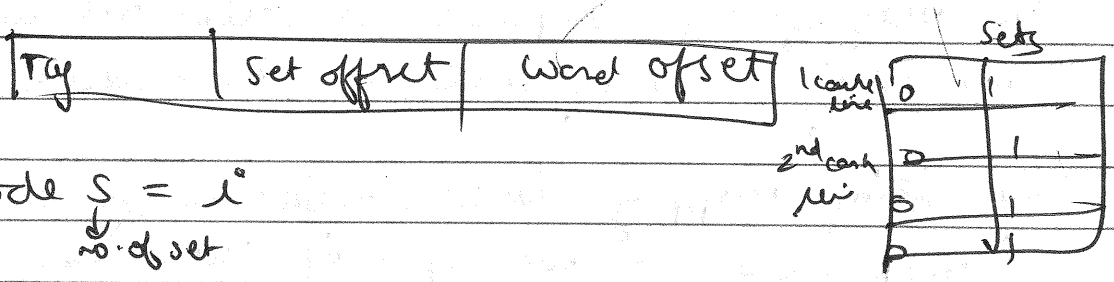
Cache size = tag memory size + data memory size.

Data memory size = no. of cache lines × no. of bits in line

(iii) set associative mapping

Cache memory is organised into sets, each set is capable to ~~hold~~ hold multiple main memory block.

no. of sets = $\frac{N}{P\text{-way}}$ → cache lines
→ no of mm blocks possible



k mode S = i
↓
no. of set

Max of tag Comparision = N

When P = 1, set associative is direct mapping

Implied / Implicit → STC, CLC, ADD

Immediate → MOV, r_0, #23

direct / absolute → $r_0 \leftarrow r_0 +$ M[2000]
ADD, A, B

Indirect → reg indirect — add, r_0, @r_1
→ mem indirect — add, r_0, @2000

indexed — ADD, r_0, 3(r_1) or ([base rg] + [index])

auto indexed — MOV r_0, +(r_1) or −
(r_1)+

indirect indexed

relative / PC
PC ← PC + displacement

base reg Am.
PC ← base reg + displacement.